AT START OF DIFFERENTIAL EQUATIONS QUESTION THERE IS SOMETIMES A PROOF. THIS PROOF HAS NOTHING TO DO WITH DIFFERENTIAL EQUATIONS

TYPE 1: SIMPLE STATEMENTS..

5 A certain substance is formed in a chemical reaction. The mass of substance formed t seconds after the start of the reaction is x grams. At any time the rate of formation of the substance is proportional to $(20 - x)$. When $t = 0$, $x = 0$ and $\frac{dx}{dt} = 1$.

(i) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0.05(20 - x). \qquad [2]$$

Rate of x proportional $(20-x)$

$$\frac{dx}{dt} \propto (20-x)$$

$$\frac{dx}{dt} = k(20-x)$$

$$1 = k(20-0)$$

$$1 = 20k$$

$$k = 0.05$$

$$\boxed{\frac{dx}{dt} = 0.05(20-x)}$$

8 In a chemical reaction, a compound X is formed from two compounds Y and Z. The masses in grams of X, Y and Z present at time t seconds after the start of the reaction are x, $10 - x$ and $20 - x$ respectively. At any time the rate of formation of X is proportional to the product of the masses of Y and Z present at the time. When $t = 0$, $x = 0$ and $\frac{dx}{dt} = 2$.

(i) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0.01(10 - x)(20 - x).$$ [1]

X	Y	Z
x	$10 - x$	$20 - x$

Rate of formation of x proportional to Product of Y and Z

$$\frac{dx}{dt} \propto \overset{Y}{(10-x)}\overset{Z}{(20-x)}$$

$$\frac{dx}{dt} = k(10-x)(20-x)$$

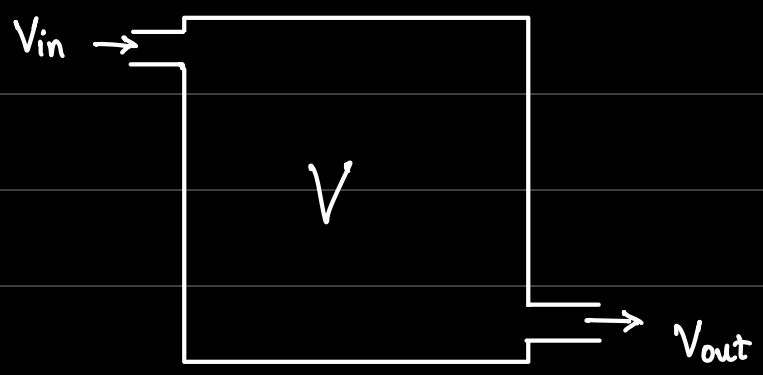
$$2 = k(10-0)(20-0)$$

$$2 = 200k$$

$$k = 0.01$$

$$\frac{dx}{dt} = 0.01(10-x)(20-x)$$

TYPE 2 :



$$V = V_{in} - V_{out}$$

$$\frac{dV}{dt} = \frac{dV_{in}}{dt} - \frac{dV_{out}}{dt}$$

Rate of change of volume = Rate of change of inflow − Rate of change of outflow.

2 In a certain industrial process, a substance is being produced in a container. The mass of the substance in the container t minutes after the start of the process is x grams. At any time, the rate of formation of the substance is proportional to its mass. Also, throughout the process, the substance is removed from the container at a constant rate of 25 grams per minute. When $t = 0$, $x = 1000$ and $\frac{dx}{dt} = 75$.

(i) Show that x and t satisfy the differential equation

$$\frac{dx}{dt} = 0.1(x - 250).$$ [2]

x

FORMATION

x_{in}

REMOVED

x_{out}

Rate of formation of x $\propto$ its mass (x)

$$\boxed{\frac{dx_{out}}{dt} = 25}$$

$$\frac{dx_{in}}{dt} \propto x$$

$$\boxed{\frac{dx_{in}}{dt} = Kx}$$

$$\frac{dx}{dt} = \frac{dx_{in}}{dt} - \frac{dx_{out}}{dt}$$

$$\boxed{\frac{dx}{dt} = Kx - 25}$$

$$75 = K(1000) - 25$$

$$100 = K(1000)$$

$$K = 0.1$$

$$\frac{dx}{dt} = 0.1x - 25$$

$$\boxed{\frac{dx}{dt} = 0.1(x - 250)}$$

TYPE 3: CHAIN RULE (LHS of proof and our equation will not be same).

11 In a model of the expansion of a sphere of radius r cm, it is assumed that, at time t seconds after the start, the rate of increase of the surface area of the sphere is proportional to its volume. When $t = 0$, $r = 5$ and $\frac{dr}{dt} = 2$.

(i) Show that r satisfies the differential equation

$$\frac{dr}{dt} = 0.08r^2. \qquad [4]$$

[The surface area A and volume V of a sphere of radius r are given by the formulae $A = 4\pi r^2$, $V = \frac{4}{3}\pi r^3$.]

Rate of increase of Surface area (A) $\propto$ Volume (V)

$$\frac{dA}{dt} \propto V$$

$$\frac{dA}{dt} \propto \frac{4}{3}\pi r^3$$

$$\frac{dA}{dt} = K\left[\frac{4}{3}\pi r^3\right]$$

$$\frac{dr}{dt} \times 8\pi r = K\left[\frac{4}{3}\pi r^3\right]$$

$$\frac{dr}{dt} = K\left[\frac{\cancel{4}^{1}}{3}\cancel{\pi} r^{\cancel{3}\,2}\right] \times \frac{1}{\cancel{8}_{2}\cancel{\pi}\cancel{r}}$$

$$\frac{dr}{dt} = K\left(\frac{r^2}{6}\right)$$

A $\qquad$ r $\qquad$ $A = 4\pi r^2$

$$\frac{dA}{dt} \qquad \frac{dr}{dt} \qquad \frac{dA}{dr} = 8\pi r$$

$$\frac{dA}{dt} = \frac{dr}{dt} \times \frac{dA}{dr}$$

$$\frac{dA}{dt} = \frac{dr}{dt} \times 8\pi r$$

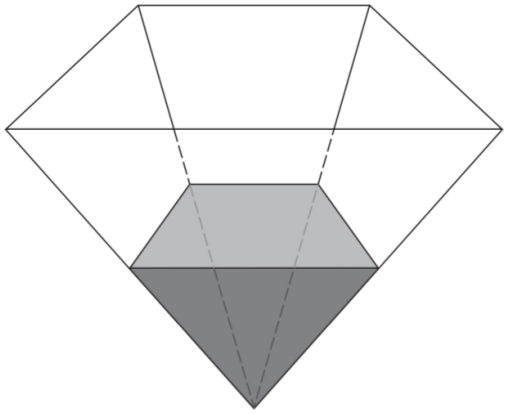
$$2 = K\left(\frac{5^2}{6}\right)$$

$$K = \frac{12}{25} = 0.48$$

$$\frac{dh}{dt} = 0.48\left(\frac{h^2}{6}\right)$$

$$\boxed{\frac{dh}{dt} = 0.08h^2}$$

9



An underground storage tank is being filled with liquid as shown in the diagram. Initially the tank is empty. At time t hours after filling begins, the volume of liquid is $V\,\mathrm{m}^3$ and the depth of liquid is $h\,\mathrm{m}$. It is given that $V = \frac{4}{3}h^3$.

The liquid is poured in at a rate of $20\,\mathrm{m}^3$ per hour, but owing to leakage, liquid is lost at a rate proportional to h^2. When $h = 1$, $\dfrac{\mathrm{d}h}{\mathrm{d}t} = 4.95$.

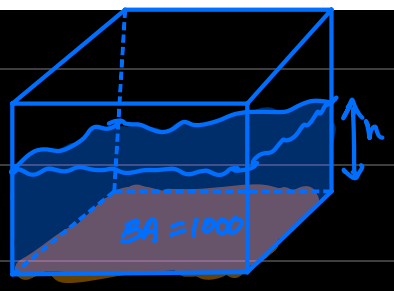
(i) Show that h satisfies the differential equation

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{5}{h^2} - \frac{1}{20}.$$ [4]

1 A rectangular reservoir has a horizontal base of area $1000\,\mathrm{m}^2$. At time $t = 0$, it is empty and water begins to flow into it at a constant rate of $30\,\mathrm{m}^3\,\mathrm{s}^{-1}$. At the same time, water begins to flow out at a rate proportional to $\sqrt{h}$, where h m is the depth of the water at time t s. When $h = 1$, $\frac{\mathrm{d}h}{\mathrm{d}t} = 0.02$.

(i) Show that h satisfies the differential equation

$$\frac{\mathrm{d}h}{\mathrm{d}t} = 0.01(3 - \sqrt{h}). \qquad [3]$$



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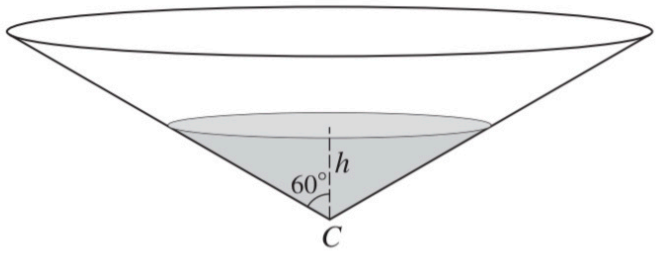

A tank containing water is in the form of a cone with vertex C. The axis is vertical and the semi-vertical angle is $60°$, as shown in the diagram. At time $t = 0$, the tank is full and the depth of water is H. At this instant, a tap at C is opened and water begins to flow out. The volume of water in the tank decreases at a rate proportional to $\sqrt{h}$, where h is the depth of water at time t. The tank becomes empty when $t = 60$.

(i) Show that h and t satisfy a differential equation of the form

$$\frac{dh}{dt} = -Ah^{-\frac{3}{2}},$$

where A is a positive constant. [4]